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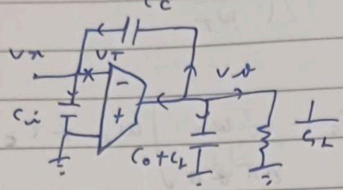
Roll number	EP22B020

5.1:

11.3.26 Assignment - 5

A1. a)



$(v_o - v_x) sC_c = v_x \times sC_i$

$\therefore v_o = v_x \left(\frac{C_i + C_c}{C_c}\right)$

$G_{m2} v_T + (v_o - v_x) sC_c + v_o[sC_o + sC_L + G_L] = 0$

$\therefore G_{m2} v_T = -v_x\left[sC_i + \left(\frac{C_i+C_c}{C_c}\right)(sC_o + sC_L + G_L)\right]$

$\therefore \frac{v_x}{v_T} = LG_2(s) = \frac{-G_{m2} C_c}{s[C_iC_c + (C_i+C_c)(C_o+C_L)] + G_L(C_i+C_c)}$

→ Pole of the closed loop transfer function, $f_2 = \omega_{u,loop}$ of $LG_2(s)$

$LG_2(s) = \frac{A_o}{1+\frac{s}{a}}$

$|LG_2(s)| = 1 \Rightarrow \frac{A_o^2}{1+\left(\frac{\omega_{u,loop_2}}{a}\right)^2} = 1$

$\therefore (\omega_{u,loop_2})^2 = a^2(A_o^2 - 1) \simeq a^2 A_o^2$

$\therefore \omega_{u,loop_2} = f_2 = aA_o = \frac{G_L(C_i+C_c)}{C_cC_i + (C_i+C_c)(C_o+C_L)} \times \frac{G_{m2}C_c}{G_L(C_i+C_c)}$

$\therefore f_2 = \frac{G_{m2}C_c}{C_cC_i + (C_i+C_c)(C_o+C_L)} = 4\times \omega_{u,loop}$

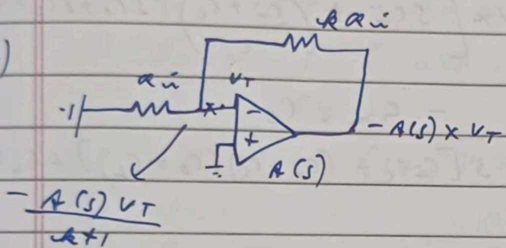
$$\therefore \frac{G_{m2} C_c}{C_c \times \frac{G_{m2}}{\omega_T} + \left(\frac{G_{m2}}{\omega_T} + C_c\right)\left(\frac{G_{m2}}{\omega_T} + C_L\right)} = 8\pi f_B \qquad (\because \omega_{u,loop} \simeq 2\pi f_B)$$

$$\therefore G_{m2}^2 + G_{m2}\left[C_L + 2C_c - \frac{C_c \omega_T}{8\pi f_B}\right]\omega_T + C_c C_L \omega_T^2 = 0$$

$$\therefore G_{m2}^2 - 27.577 \times 10^{-2} G_{m2} + 96 \times 10^{-4} = 0$$

$$\therefore G_{m2} = 0.040875,\ 0.23495$$

b)



$$\therefore L_G(s) = -\frac{A(s)}{k+1}$$

Assume $A(s) = \frac{(1 - s/z_1) A_0}{\left(1 + \frac{s}{p_1}\right)\left(1 + \frac{s}{p_2}\right)}$

$$\therefore \omega_{u,loop} :- |L_G(s)| = 1 \Rightarrow 1 = \left(\frac{A_0}{k+1}\right)^2 \frac{\left[1 + \omega_u^2/z_1^2\right]}{\left[1 + \omega_u^2/p_1^2\right]\left[1 + \omega_u^2/p_2^2\right]}$$

- Assuming $\omega_u \ll p_2, z_1$ and $\omega_u \gg p_1$:-

$$1 = \left(\frac{A_0}{k+1}\right)^2 \times \frac{1}{\omega_u^2/p_1^2} \Rightarrow \omega_u = \frac{A_0 p_1}{k+1}$$

- Assuming $A_0 = \frac{G_{m1} G_{m2}}{G_{o1} G_{o2}}$ and $p_1 \simeq \frac{G_{o1} G_{o2}}{G_{m2} C_c}$

$$\omega_{u,loop} \simeq 2\pi f_B = \frac{G_{m1}}{(k+1) C_c}$$

$\therefore G_{m_1} = 2\pi f_0 (A+1) C_c = 1130.97 \times 10^{-6} S$

$G_{m_1} = 1.13\ mS$

$\rightarrow \dfrac{G_{m_1}}{C_1} = \dfrac{1.13 \times 10^{-3}}{\dfrac{1.13 \times 10^{-3}}{\omega_T} + \dfrac{G_{m_2}}{\omega_T}} = \dfrac{\omega_T G_{m_1}}{G_{m_1} + G_{m_2}} \simeq \dfrac{\omega_T G_{m_1}}{G_{m_2}}$

$\dfrac{G_{m_2}}{C_2} = \dfrac{G_{m_2}}{\dfrac{G_{m_2}}{\omega_T} + C_L} = \dfrac{\omega_T G_{m_2}}{G_{m_2} + C_L \omega_T} = \dfrac{\omega_T G_{m_2}}{G_{m_2} + 0.48}$

- For $G_{m_2} = 0.04087S$, $\dfrac{G_{m_1}}{C_1} = 0.027\omega_T$ $\left(\dfrac{G_{m_2}}{C_2} = 0.078\omega_T\right.$

For $G_{m_2} = 0.235S$, $\dfrac{G_{m_1}}{C_1} = 0.005\omega_T$ $\left(\dfrac{G_{m_2}}{C_2} = 0.33\omega_T\right.$

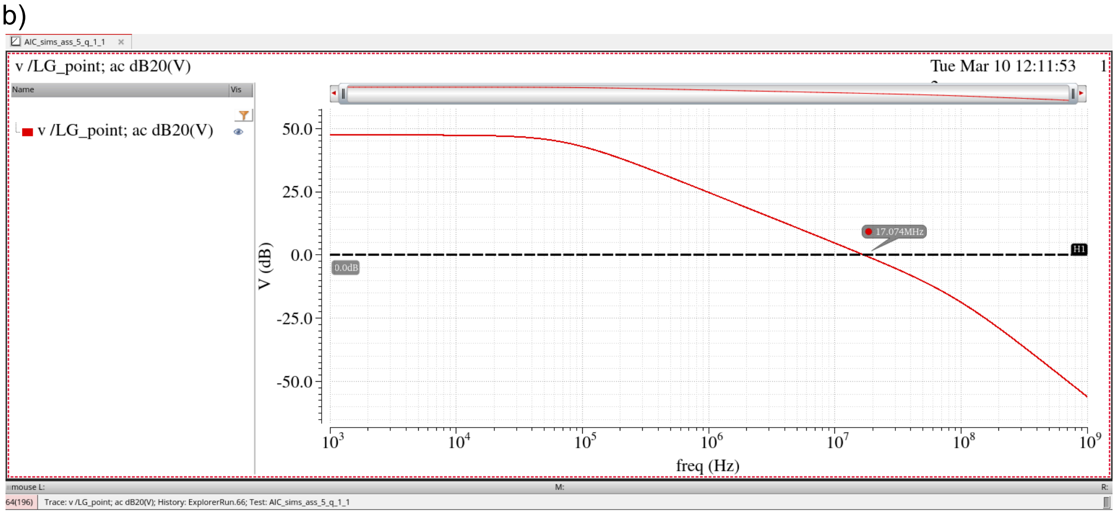
- ~~Thus, both values for G_{m_2} are permissible. $G_{m_2} = 0.235S$ gives less steady state error, since it gives a higher A_0.~~
- $G_{m_2} = 0.0408S$ has less power consumption ($\because$ power consumption $\propto G_m$, because the parasitics will be higher for higher G_m)

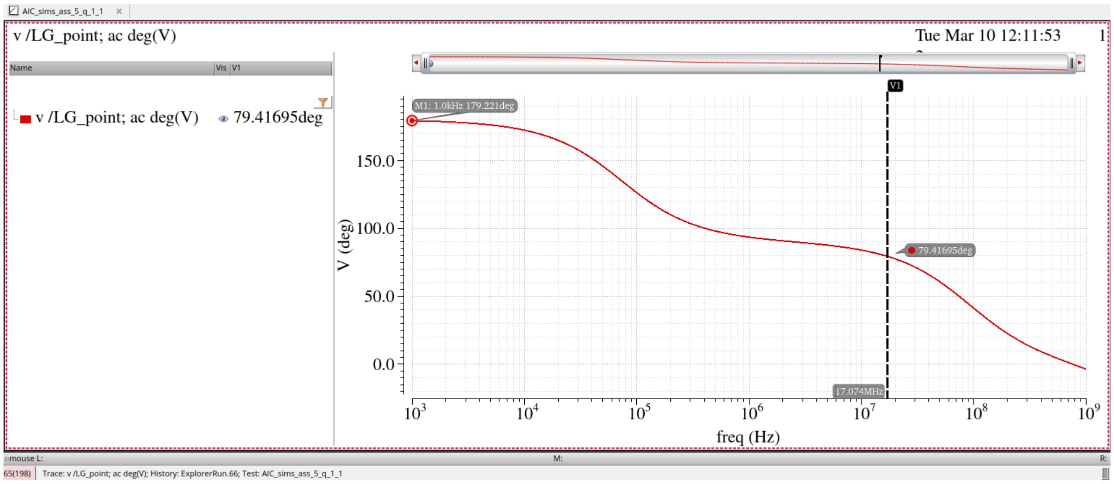
Thus, we will use Gm_2 = 0.04087S for further analysis.

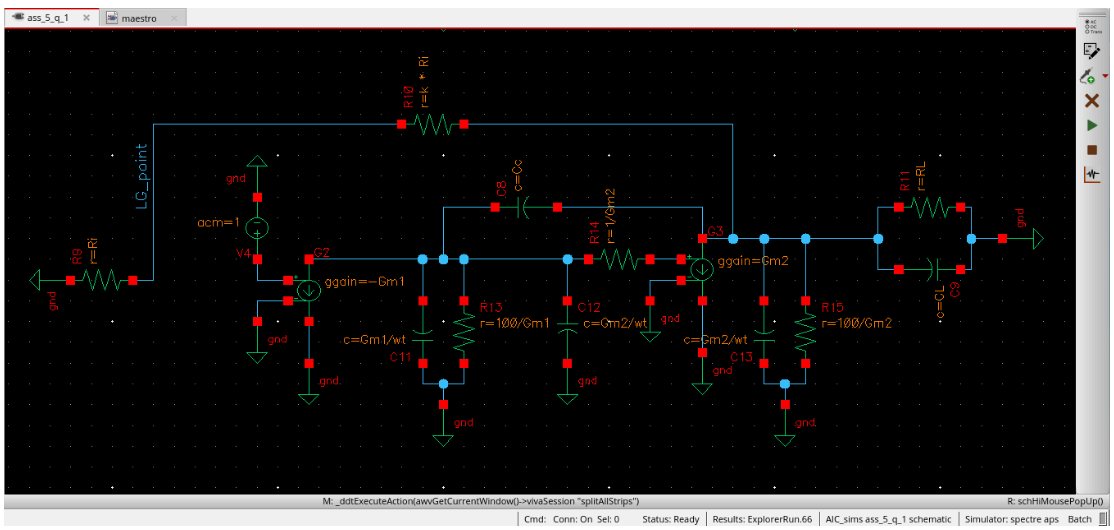
a)

Specification	Value	Unit
Closed loop dc gain k	8	-
Closed loop bandwidth f_B	20	MHz
Load capacitor C_L	24	pF
Load resistor R_L	663.14	Ω
Input resistance Ri	10	kΩ
Gm_1	1.13	mS
Gm_2	0.04087	S
$C_{i,1} = C_{o,1} = Gm_1/\omega_T$	0.0565	pF
$C_{i,2} = C_{o,2} = Gm_2/\omega_T$	2.0435	pF
$R_{o,1} = 100/Gm_1$	88.495	kΩ
$R_{o,2} = 100/Gm_2$	2.447	kΩ
$R_{N,1} = 1/Gm_1$	884.95	Ω
$R_{N,1} = 1/Gm_2$	24.47	Ω
C_c	1	pF

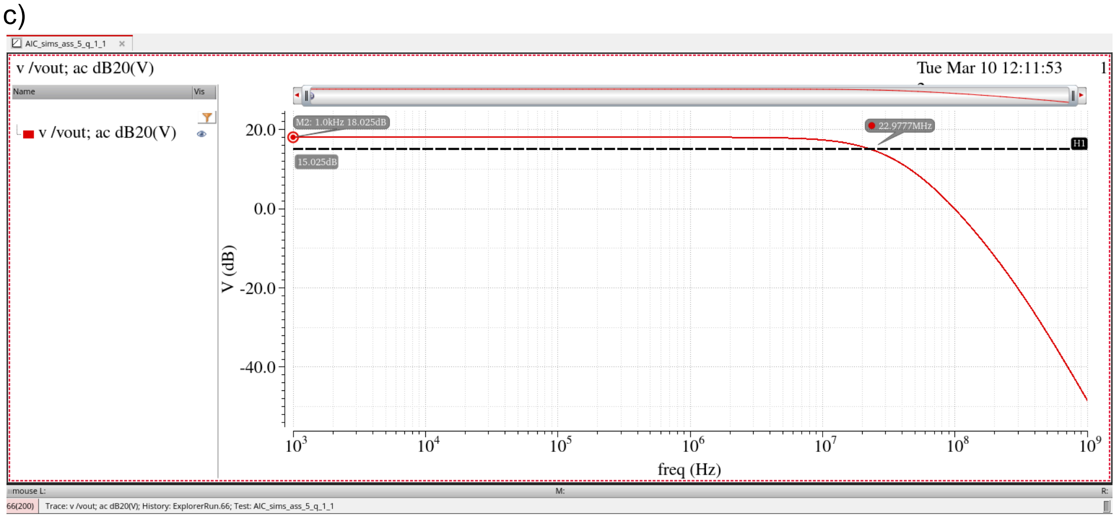
b)



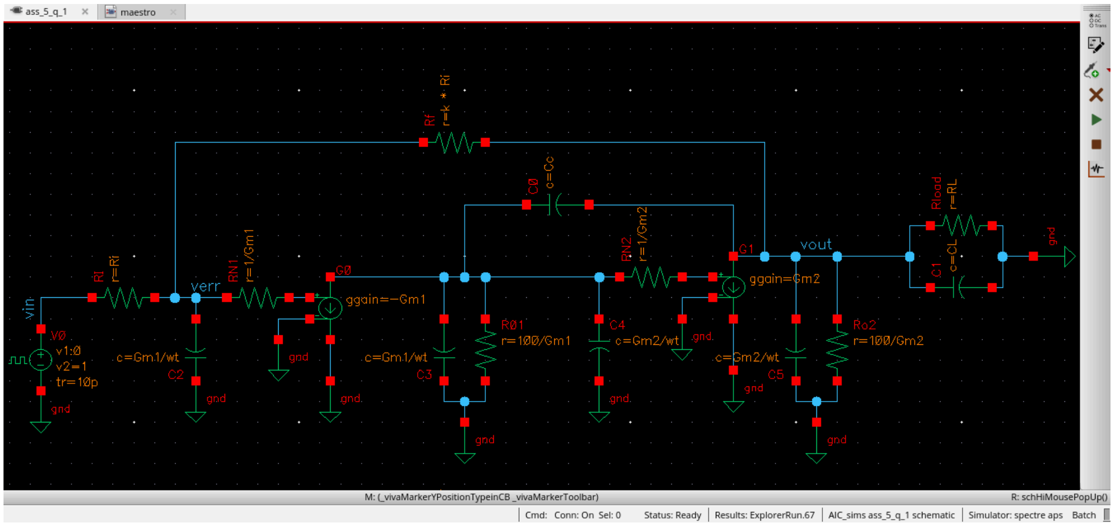




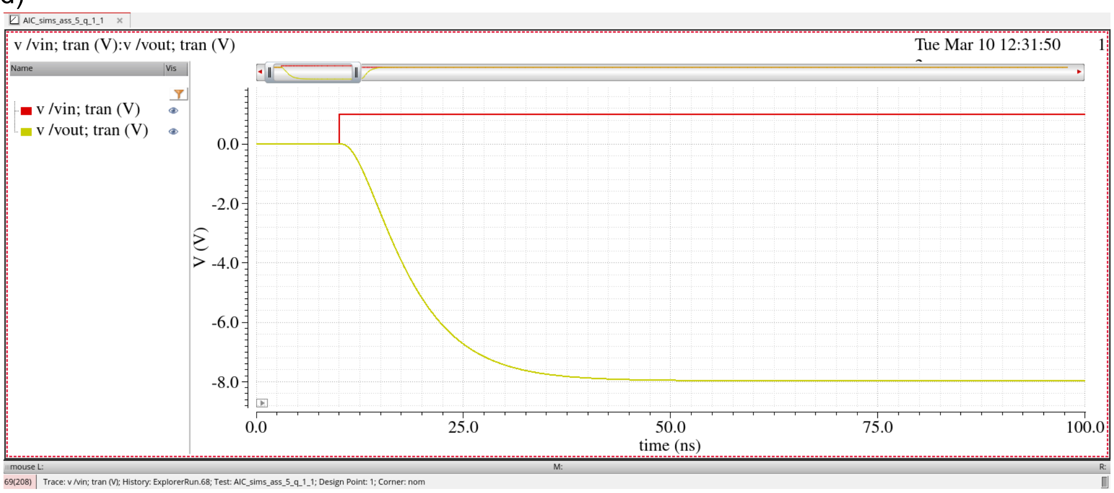
c)



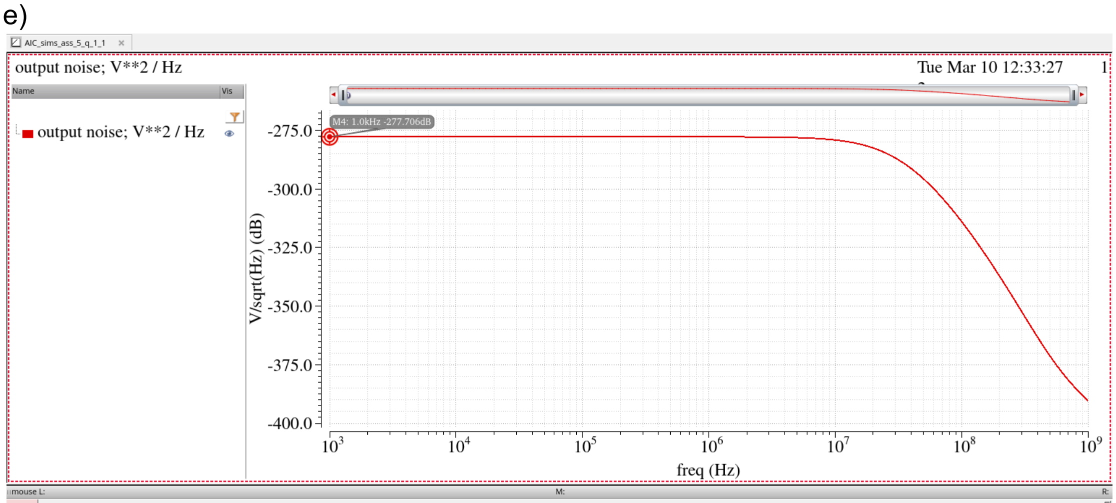
For DC gain of 8, 20*log(8) = 60 * 0.301 = 18.06dB

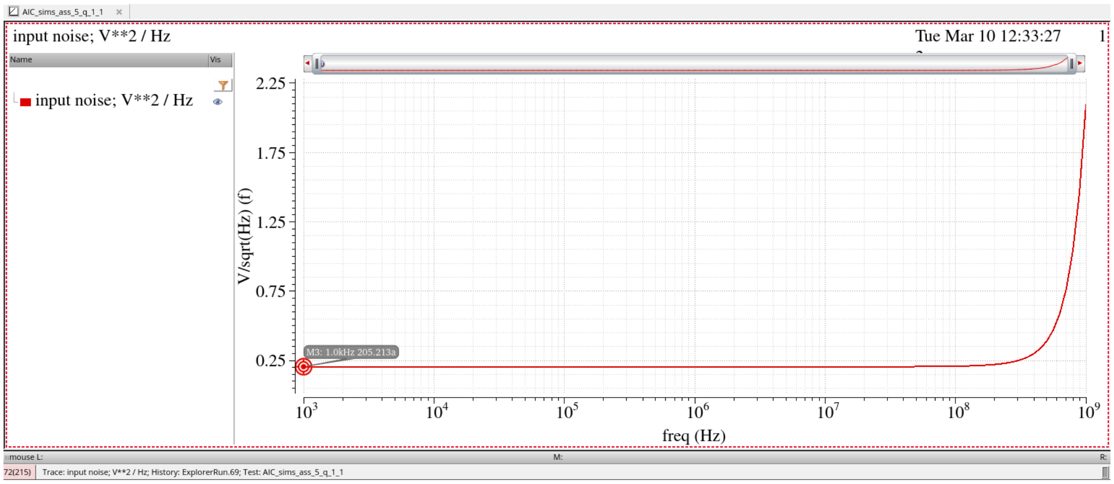

ass_5_q_1
maestro
Rf
r=k * Ri
C0
c=Cc
Rload
r=RL
Ri
r=Ri
verr
RN1
r=1/Gm1
G0
ggain=-Gm1
RN2
r=1/Gm2
G1
ggain=Gm2
vout
C1
c=CL
vin
V0
v1:0
v2=1
tr=10p
c=Gm1/wt
C2
c=Gm1/wt
C3
R01
r=100/Gm1
C4
c=Gm2/wt
c=Gm2/wt
C5
Ro2
r=100/Gm2
gnd
Status: Ready
Results: ExplorerRun.67
AIC_sims ass_5_q_1 schematic
Simulator: spectre aps

d)


AIC_sims_ass_5_q_1_1
v /vin; tran (V):v /vout; tran (V)
Tue Mar 10 12:31:50
v /vin; tran (V)
v /vout; tran (V)
V (V)
0.0
-2.0
-4.0
-6.0
-8.0
0.0
25.0
50.0
75.0
100.0
time (ns)
Trace: v /vin; tran (V); History: ExplorerRun.68; Test: AIC_sims_ass_5_q_1_1; Design Point: 1; Corner: nom

e)

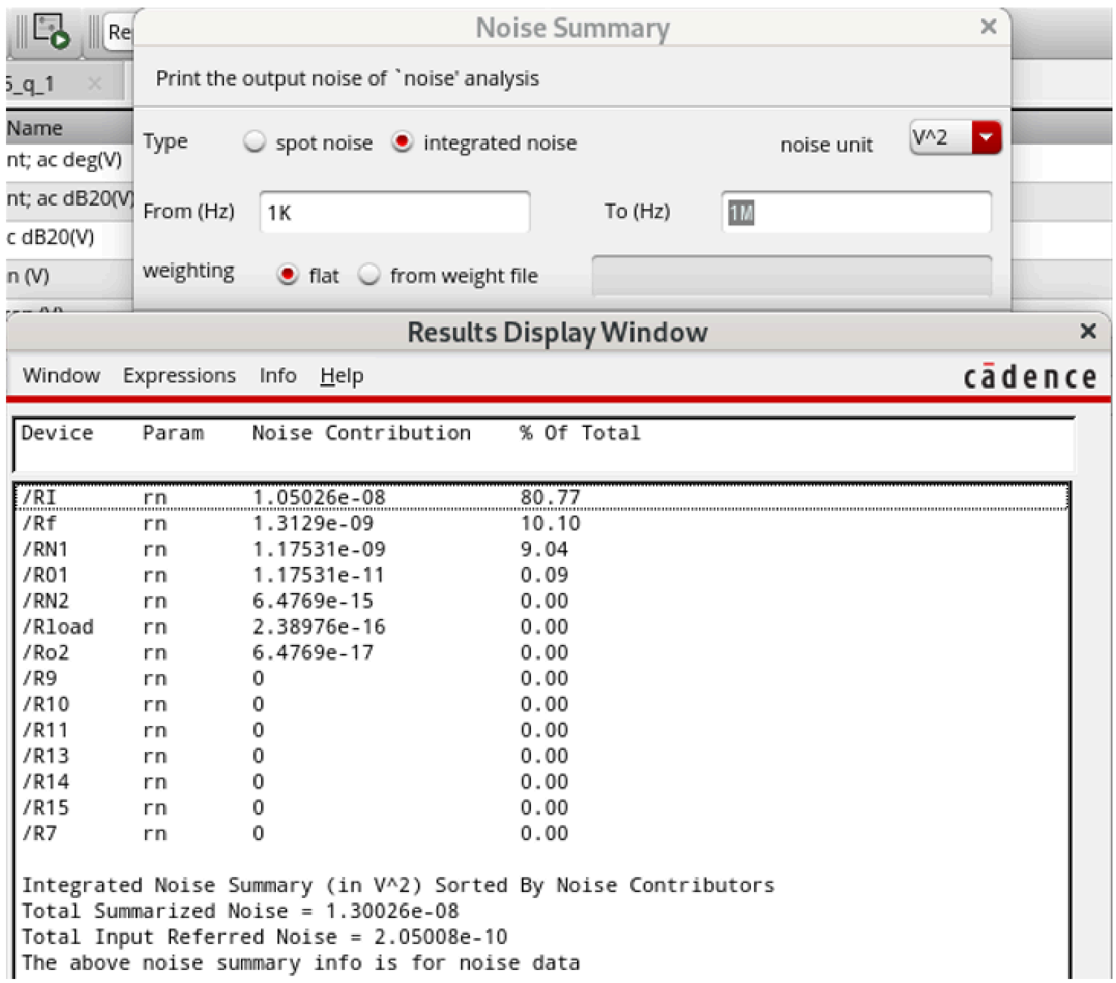

AIC_sims_ass_5_q_1_1
output noise; V**2 / Hz
Tue Mar 10 12:33:27
output noise; V**2 / Hz
M4: 1.0kHz -277.706dB
V/sqrt(Hz) (dB)
-275.0
-300.0
-325.0
-350.0
-375.0
-400.0
10^3
10^4
10^5
10^6
10^7
10^8
10^9
freq (Hz)
Trace: output noise; V**2 / Hz; History: ExplorerRun.69; Test: AIC_sims_ass_5_q_1_1



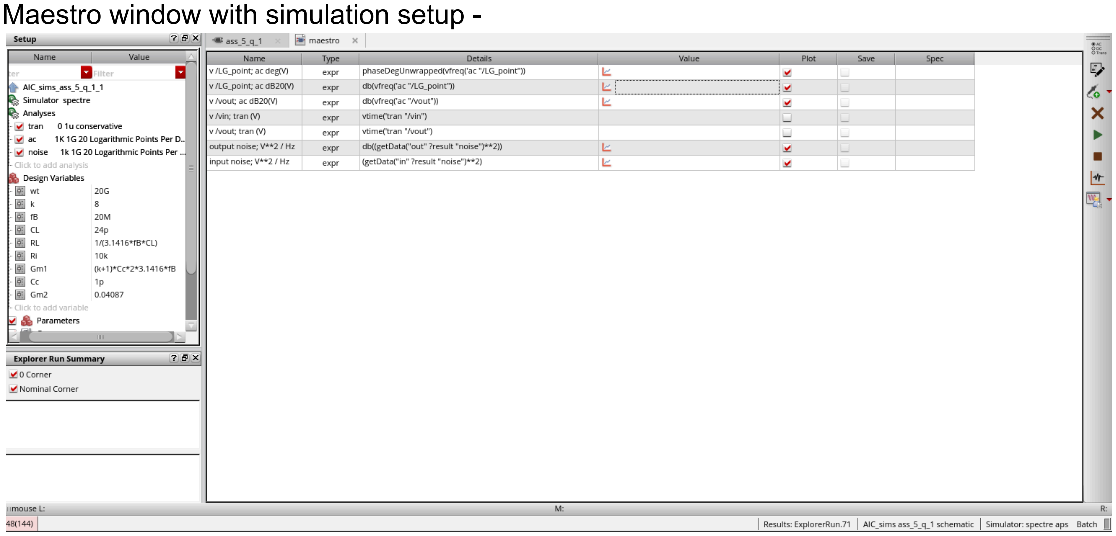
f)

Specification	Value
DC gain	18.025 dB = 7.966
3-dB bandwidth, f_B	22.98 MHz
Unity loop gain frequency	17.07 MHz
Phase margin	79.42^0
Input Noise PSD (at low frequencies)	205.21×10^{-18} V^2/Hz

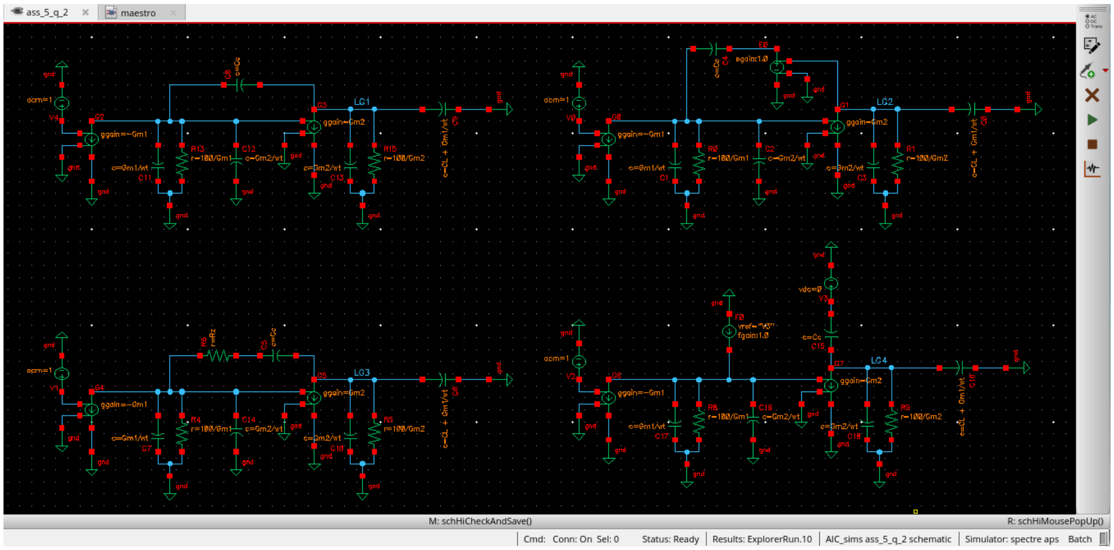
Component	% of noise contribution (low frequency)
Ri	80.77
Rf	10.10
Gm_1	09.04
Gm_2	00.00
R_L	00.00



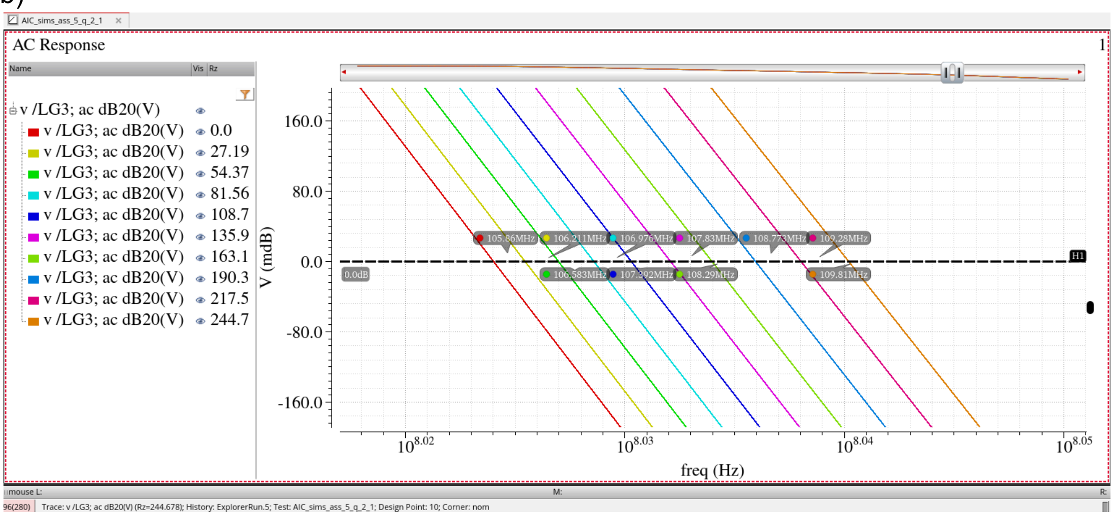
Maestro window with simulation setup -

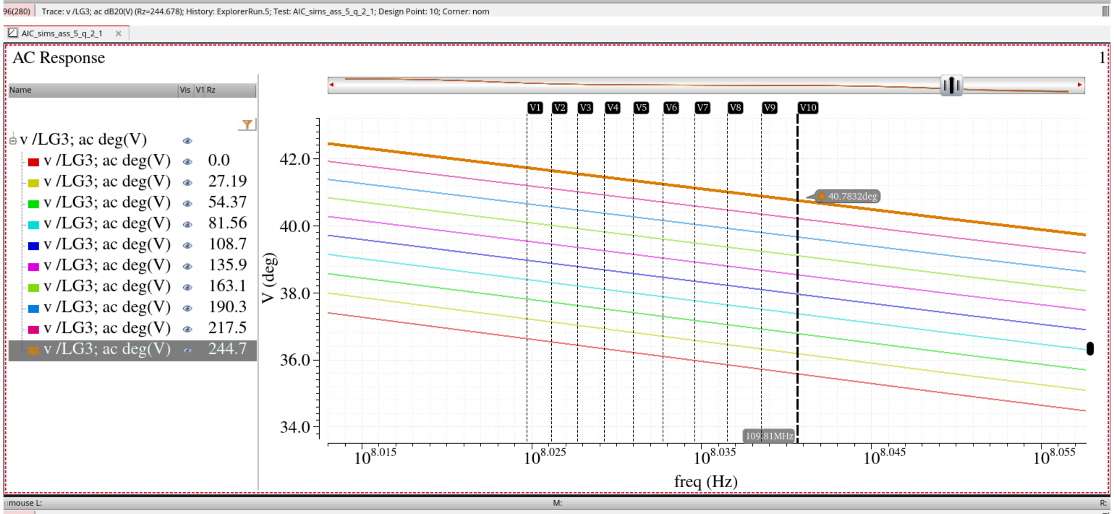


5.2:



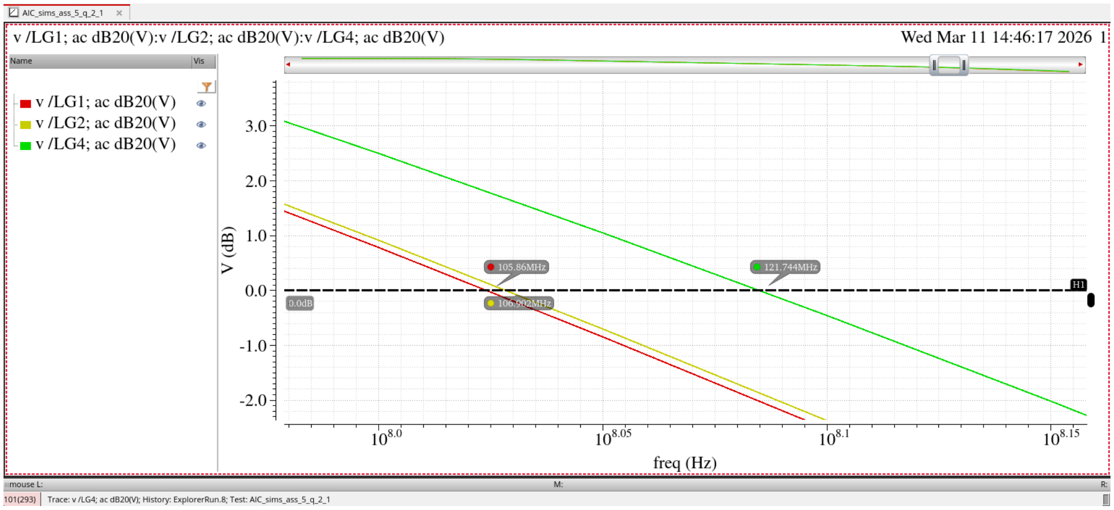
b)

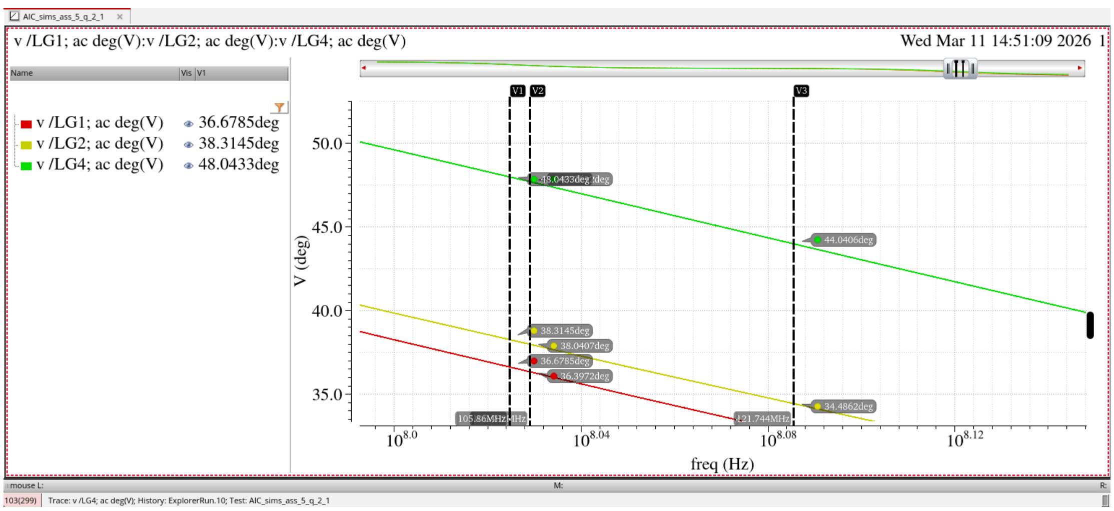




Thus, we can conclude that the higher the Rz, the higher the phase margin. This is because as Rz increases, the zero moves into the left half plane and keeps increasing the phase margin.

a, c, d)

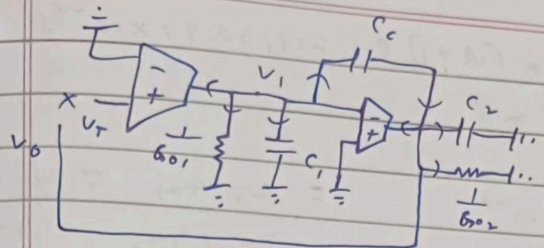




The phase margins in c) and d) are better because the zero is removed and does not contribute to the phase margin anymore. The cross terms of Cc with C_1 and G_{01} in c) and C_2 and G_{02} in d) also disappear.

Case	Phase margin
a)	36.68^0
b)	40.78^0
c)	38.04^0
d)	44.04^0

A2. a)



$$-g_{m1} v_T + v_1 (g_{o1} + sC_1) + (v_1 - v_o) sC_c = 0$$

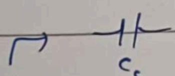
$$(v_1 - v_o) sC_c = g_{m2} v_1 + v_o (g_{o2} + sC_2)$$

$$\therefore \; v_1 = \frac{v_o (g_{o2} + sC_c + sC_2)}{sC_c - g_{m2}}$$

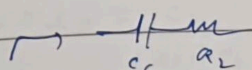
$$\therefore \; LG_1(s) = A_1(s) = \frac{v_o}{v_T} = \frac{-g_{m1}(g_{m2} - sC_c)}{s^2 (C_1 C_c + C_2 C_c + C_1 C_2) + s[g_{o1} C_2 + g_{o2} C_1 + C_c (g_{m2} + g_{o1} + g_{o2})] + g_{o1} g_{o2}}$$

b) Substituting $sC_c \rightarrow \frac{sC_c}{1 + sC_c R_2}$

$$LG_2(s) = A_2(s) = \frac{v_o}{v_T} = \frac{-g_{m1}[g_{m2} + sC_c (g_{m2} R_2 - 1)]}{s^3 C_1 C_2 R_2 C_c + s^2 [C_c C_1 + C_c C_2 + C_1 C_2 + C_c R_2 (g_{o2} C_1 + g_{o1} C_2)] + s[C_c (g_{o1} + g_{o2} + g_{m2}) + g_{o2} C_1 + g_{o1} C_2 + C_c R_2 g_{o1} g_{o2}] + g_{o1} g_{o2}}$$

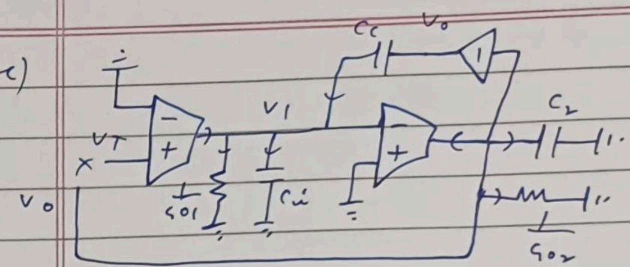


$Z_{in} = \frac{1}{sC_c}$



$Z_{in} = \frac{1 + sC_c R_2}{sC_c}$

c)

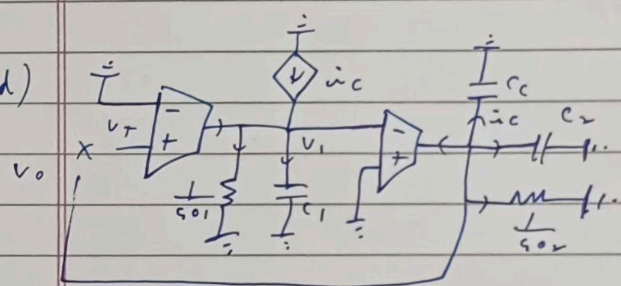


- $g_{m1} v_T + (v_o - v_1) sC_c = v_1 (g_{o1} + sC_1)$
- $g_{m2} v_1 + v_o (g_{o2} + sC_2) = 0$

$$\therefore v_1 = \frac{-v_o}{g_{m2}} (g_{o2} + sC_2)$$

- $$\therefore LG_3(s) = \frac{v_o}{v_T} = \frac{-g_{m1} g_{m2}}{s^2 (C_1 C_2 + C_c C_2) + s[g_{o1} C_2 + g_{o2} C_1 + C_c (g_{m2} + g_{o2})] + g_{o1} g_{o2}}$$
  $= A_3(s)$

d)



$i_c = v_o s C_c$

- $g_{m1} v_T + v_o s C_c = v_1 (g_{o1} + sC_1)$
- $g_{m2} v_1 + v_o s C_c + v_o (g_{o2} + sC_2) = 0$
- $$\therefore A_4(s) = \frac{g_{m1} g_{m2}}{g_{o1} g_{o2} + s[g_{o2} C_1 + g_{o1} C_2 + C_c (g_{m2} + g_{o1})] + s^2 (C_1 C_2 + C_1 C_c)}$$

→ c) Decouples C_c with C_1 and g_{o1} and removes the pole.
d) Decouples C_c from C_2 and g_{o2} and removes the pole.